



2026 SUMMER SCHOOL

Transforming Technology, Modern Practice, and Clinical Impact

JUNE 16–20 | UNIVERSITY OF MICHIGAN

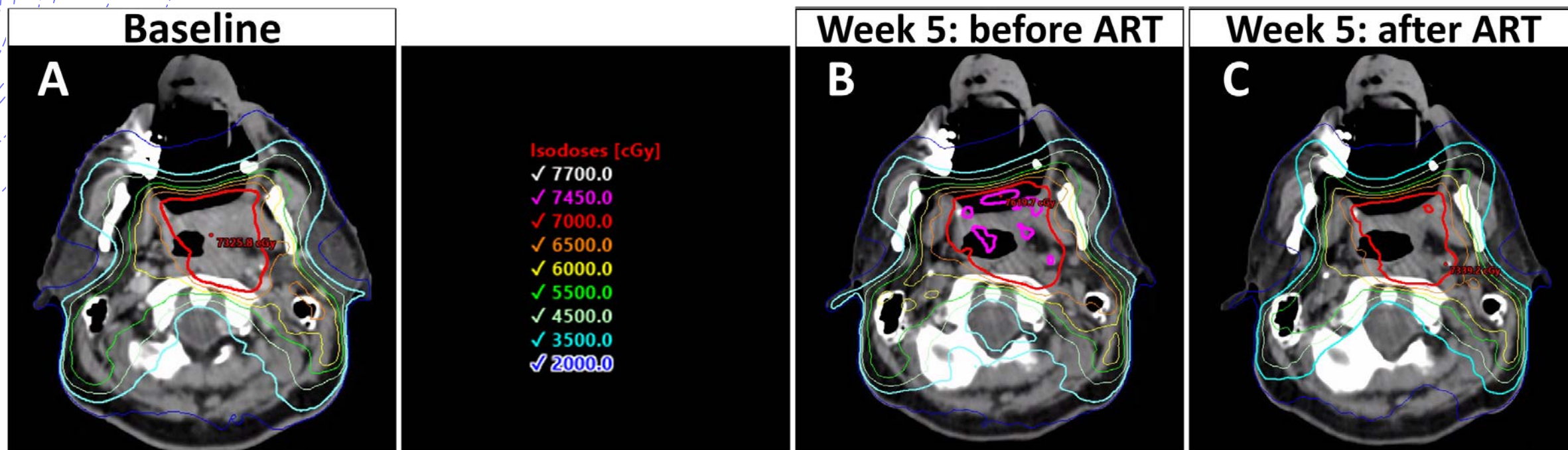
Clinical Rationale for Adaptive Radiotherapy

Dandan Zheng & Alex Podgorsak

Department of Radiation Oncology, University of Rochester,
Rochester, NY

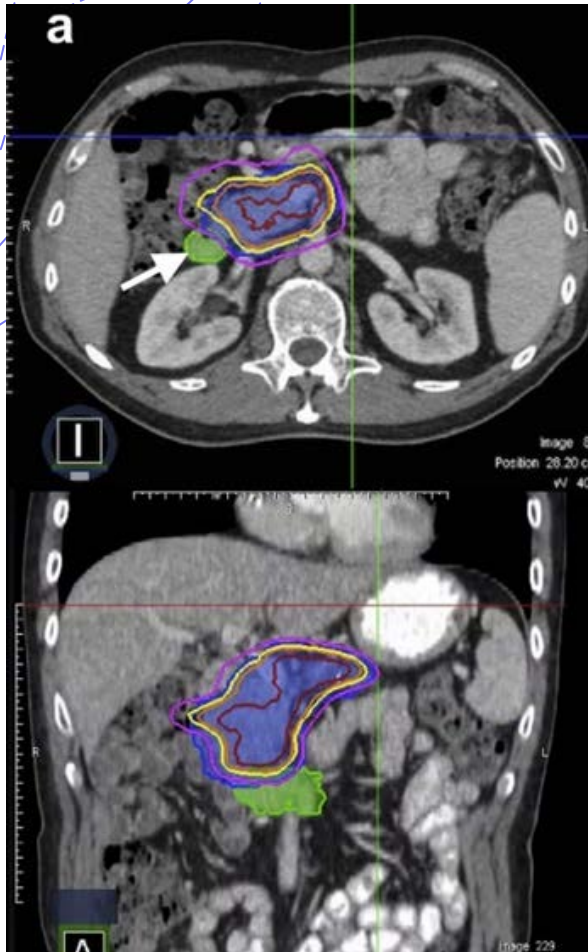


A Clinical Example

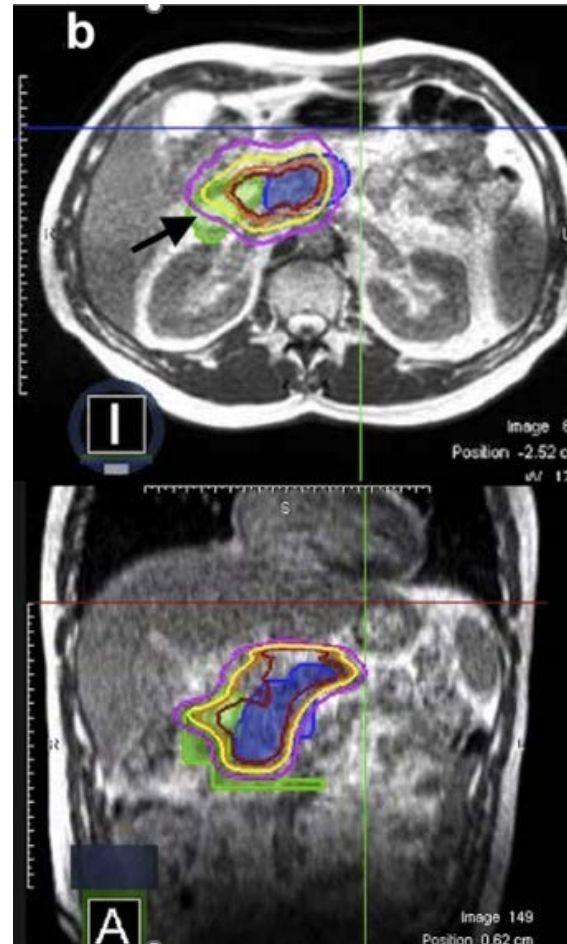


Morgan, H.E., Sher, D.J. 2020, Cancers Head Neck

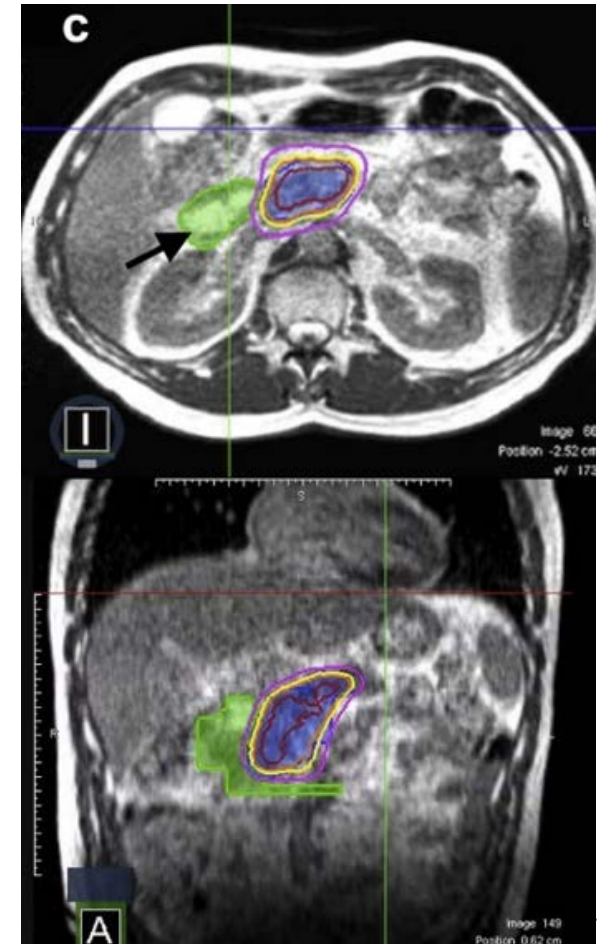
Snapshot Fallacy



Dynamic Reality



ART Solution





**What
clinical/anatomical/biological
factors, in your opinion, justify
ART?**

Snapshot Fallacy

- **Plan based on snapshot simulation**
- **Population-based static margins (e.g. van Herk formula)**

Dynamic Reality

- **Tumor shrinkage**
- **Weight loss**
- **Organ motion & deformation**

ART Solution

- **Reoptimized to new anatomy**
- **Offline; Online; Real-time**
- **Adapt to biology**

Anatomical Reality: Inter- and Intra-fractional Change

TUMOR SHRINKAGE



HN, lung, cervix...

WEIGHT LOSS & GEOMETRY SHIFTS



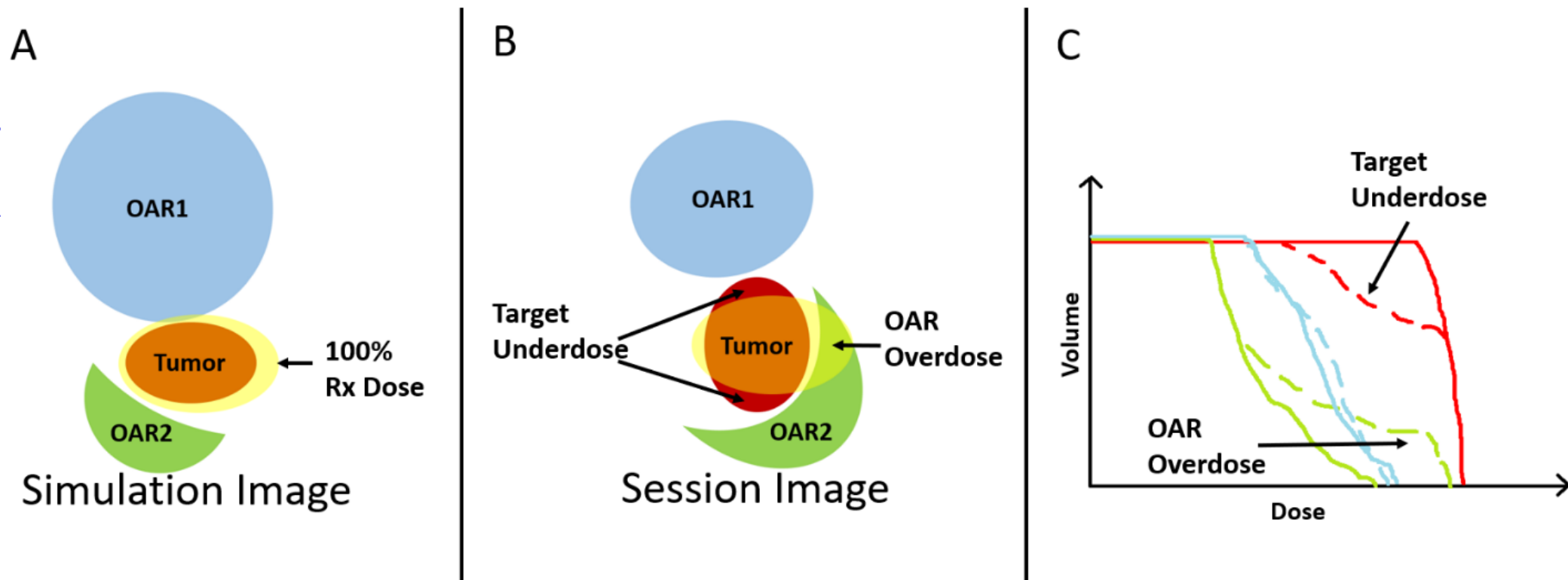
HN, GI...

DAILY ORGAN FILLING



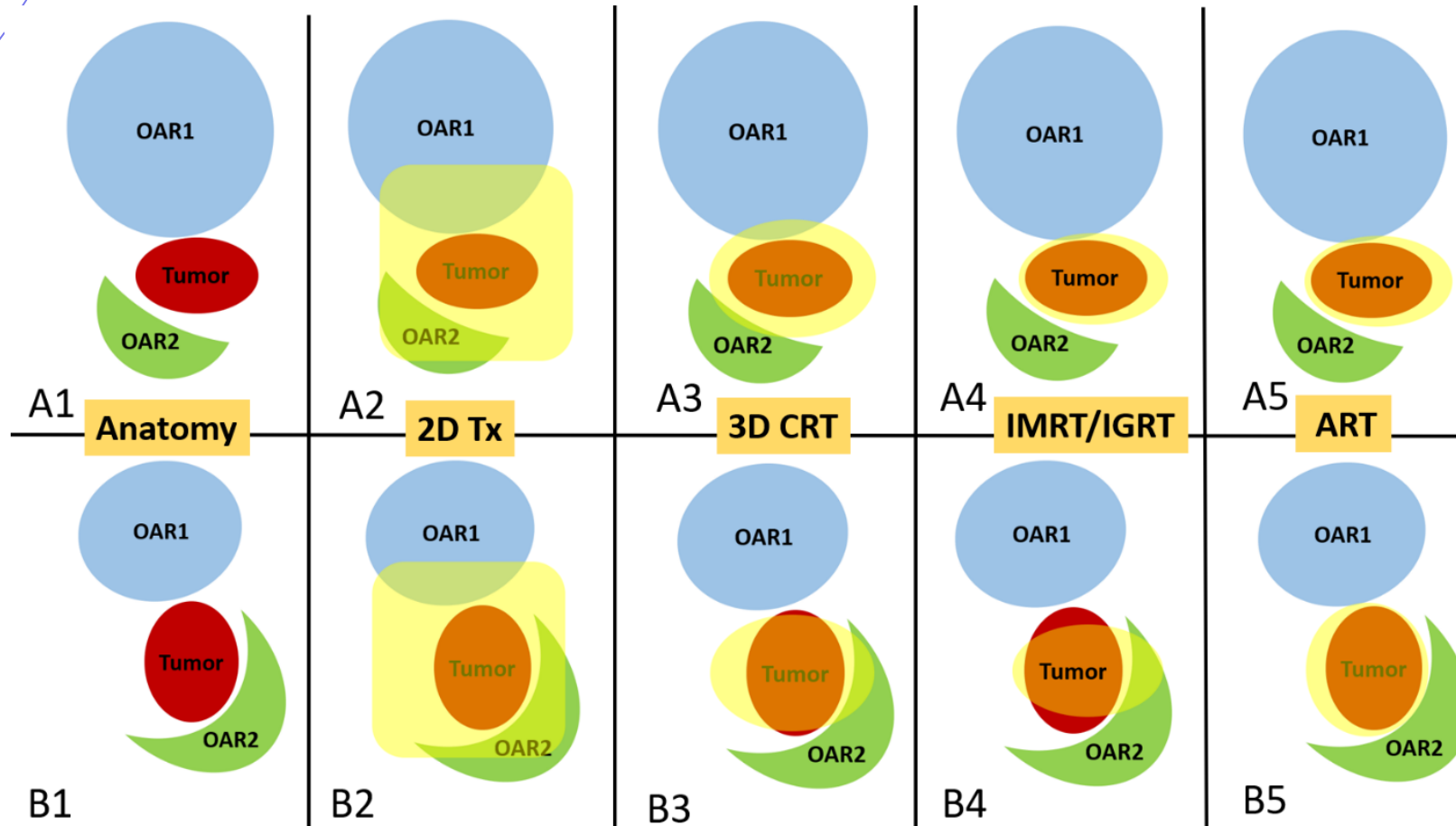
Physiological cycles
(respiration,
digestion,
stomach/bladder
/ rectum filling)

Dosimetric Fidelity



What's on paper $\neq$ What's delivered
Can lead to increased toxicity to OARs and reduced tumor control

Why Now? Irony of Precision



Dona Lemus et al. (2024) Cancers



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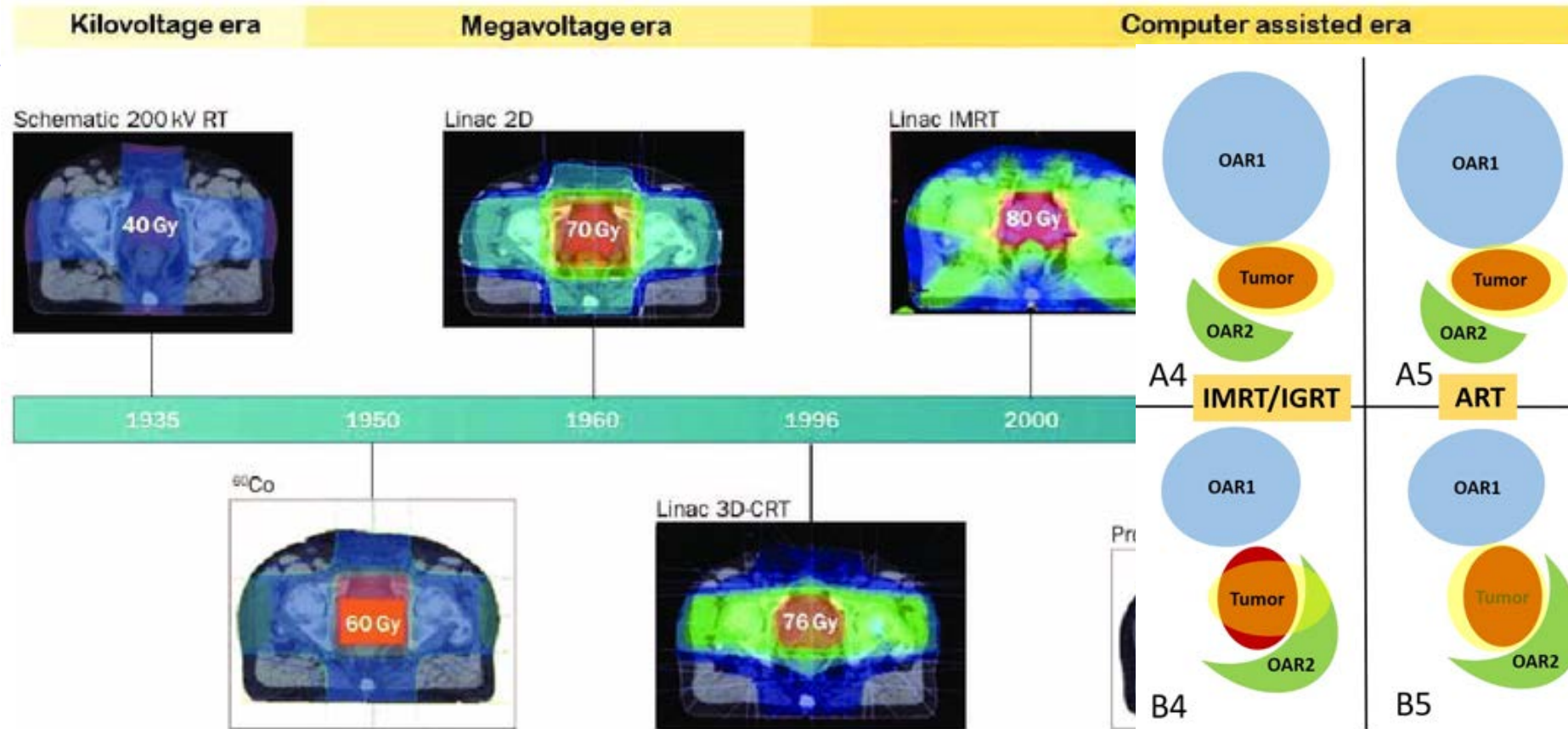


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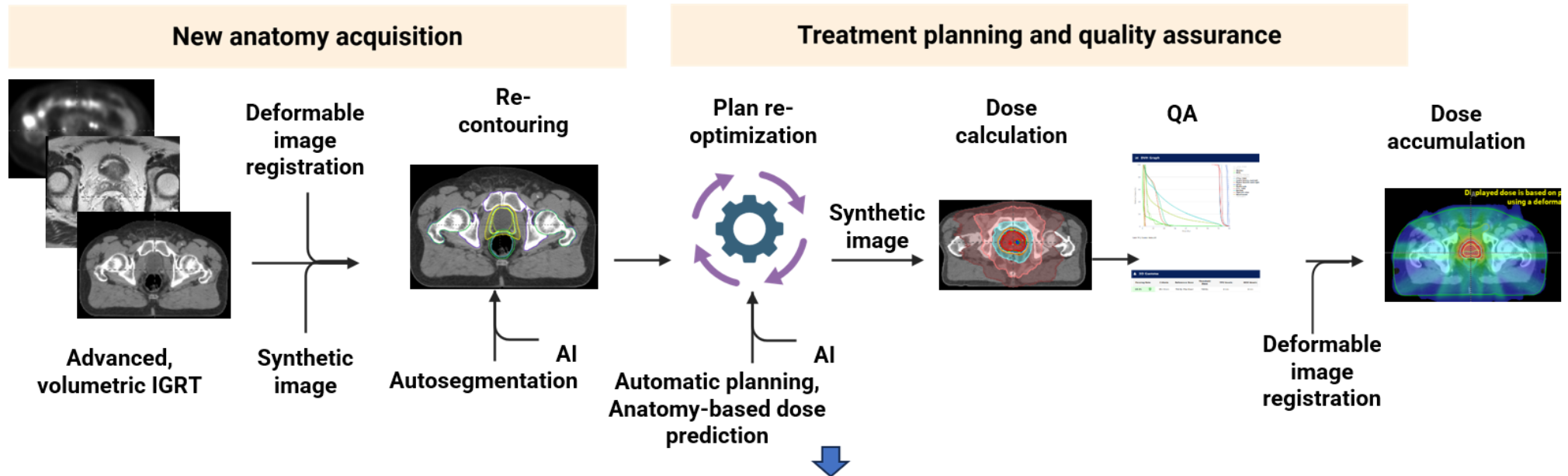
Particle Therapy Needs ART Even More



Thariat, Juliette, et al. (2013) Nat Rev Clin Oncol

Technical Rationale

Technological Enablers of Adaptive Radiotherapy



Integrated and streamlined systems



Strategic Advantages of ART



DOSE FIDELITY

Ensures the radiation delivered precisely matches the intended 'paper' plan.



MARGIN REDUCTION

Safely shrinks PTV margins, treating less normal tissue and lowering toxicity.

Saved to this PC



DOSE ESCALATION

Enables curative high-dosing in toxicity-limited areas (e.g., pancreatic cancer).



HYPOFRACTIONATION

Makes high-dose SBRT safer by protecting organs at risk despite massive per-fraction doses.



DIRECT-TO-UNIT

Enables immediate, simulation-free palliative and abdominal SBRT (STAT-ART, FAST-Mets).



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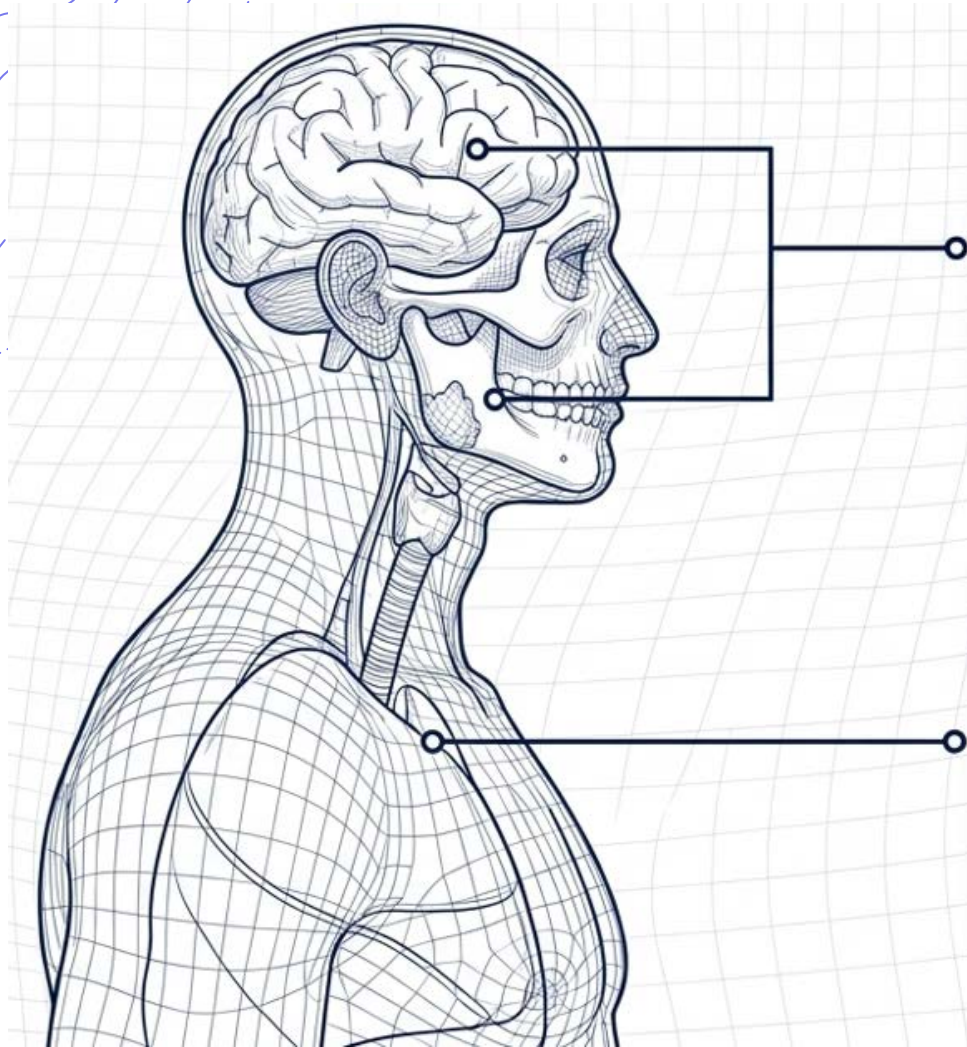
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Clinical Evidence: HN & Thorax



HEAD AND NECK (BRAIN/SALIVARY)

Trials: ARTIX, ARTFORCE, DARTBOARD, UNITED

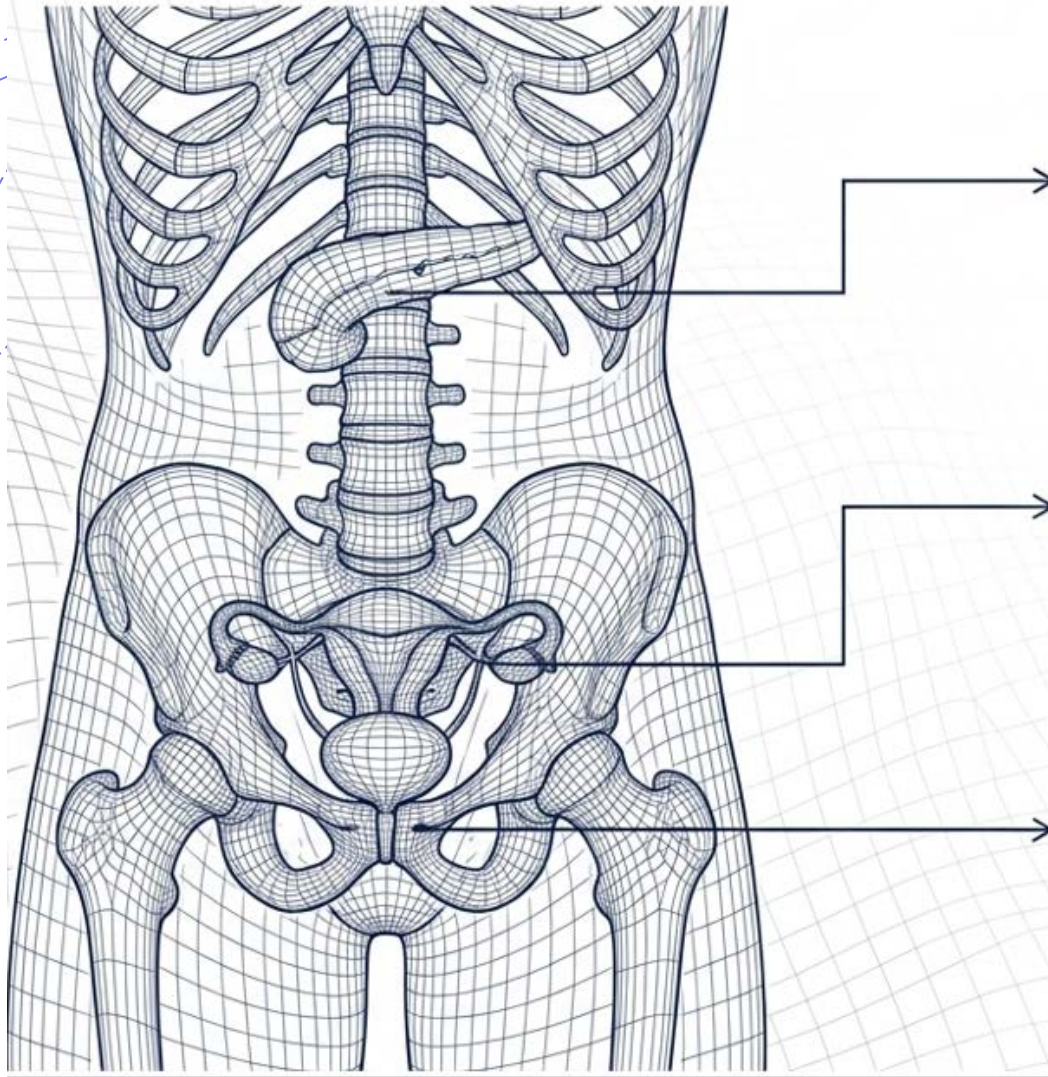
- Safely reduces mean dose to parotid glands (up to 7.1 Gy), improving excretory function.
- Allows safe margin reduction for high-grade gliomas, achieving large clinical target volume reduction.

LUNG

Trial: RTOG 1106

- Mid-treatment PET/CT-guided adaptive escalation is safe and technically feasible.
- Allows for simultaneous field reduction and dose escalation based on tumor response.

Clinical Evidence: Abdomen & Pelvis



PANCREAS

Trial: SMART

Landmark phase II trial showed daily MR-guided ART enabled substantial stomach, bowel, and duodenum sparing while achieving an overall survival benefit for unresectable disease.

CERVICAL

Improved CTV/PTV coverage while significantly reducing doses to bladder, rectum, and bowel by managing uterine motion and daily tumor shrinkage.

PROSTATE & BLADDER

Trial: RAIDER

- Safely accounts for daily bladder/rectum filling cycles.
- Allows for aggressive posterior margin reduction

Timeline of Clinical Evidence

2012–2017

RTOG 1106 (NSCLC)

Proved safety and feasibility of mid-treatment PET/CT adaptive dose escalation.

2013–2018

ARTIX (Oropharyngeal)

Offline weekly replanning improved parotid gland excretory function.

2021–2023

RAIDER (Bladder)

Demonstrated precise dose delivery in a highly mobile organ.

2012–2019

ARTFORCE (Head & Neck)

FDG-PET guided dose redistribution; safely maintained control with escalation to 84 Gy.

2021–2023

UNITED (Glioma)

Achieved large clinical target volume reduction with MR-Linac weekly adaptation.

Future

Ongoing phase III trials shifting focus from technical feasibility to long-term outcome validation.

It Is Only The Beginning...

SMART (MR)

**Anatomical
Sparing (OARs)**

Escalation via
toxicity reduction

RAIDER (CBCT)

**Anatomical
Deformation**

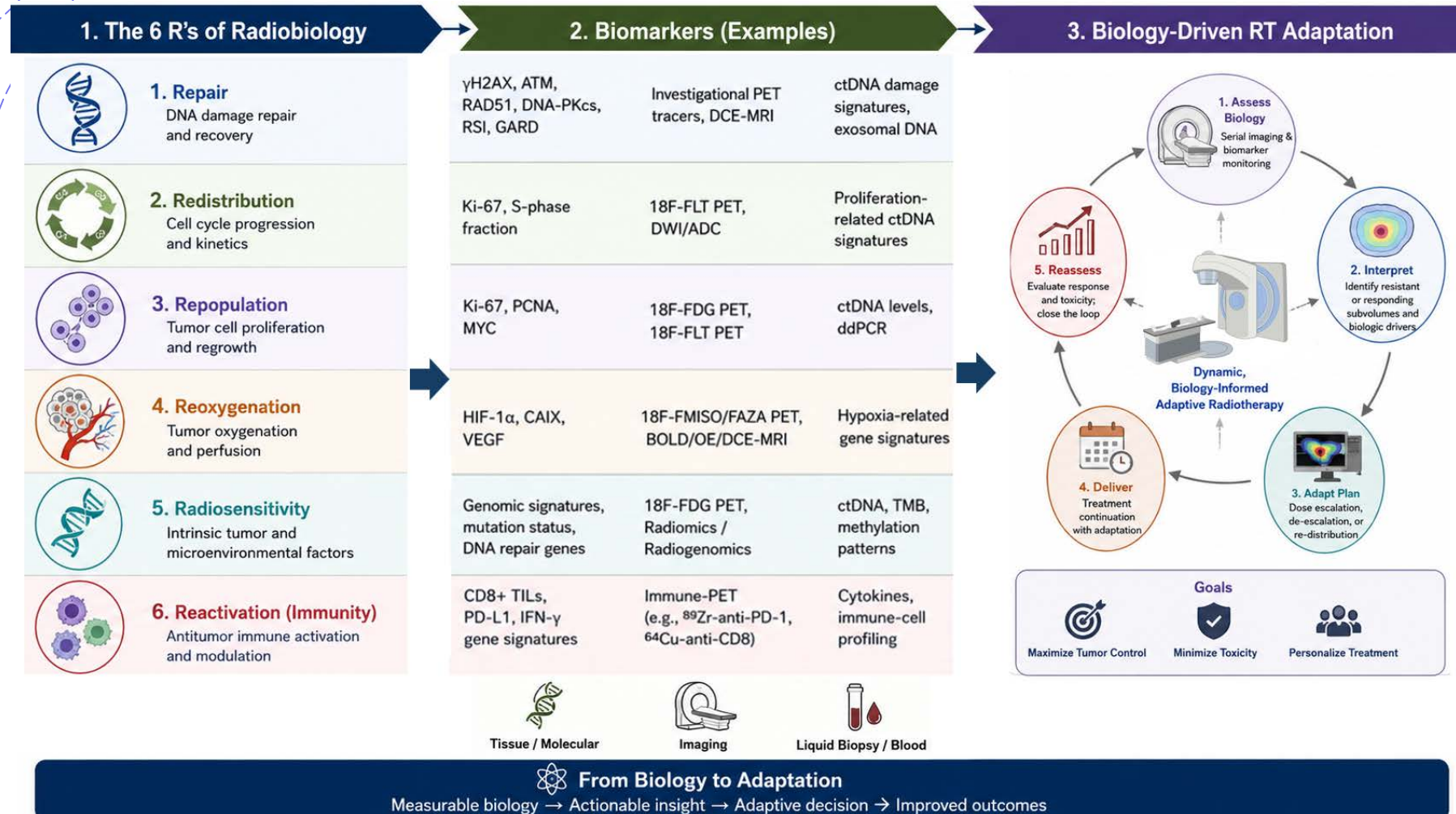
Margin reduction
& precision

ARTFORCE (PET)

**Biological
Redistribution**

Targeting
radioresistant cores

Biological Rationale and Biology-Driven ART



Delivering the Promise of Precision

1 ANATOMY IS DYNAMIC; TREATMENTS MUST BE TOO

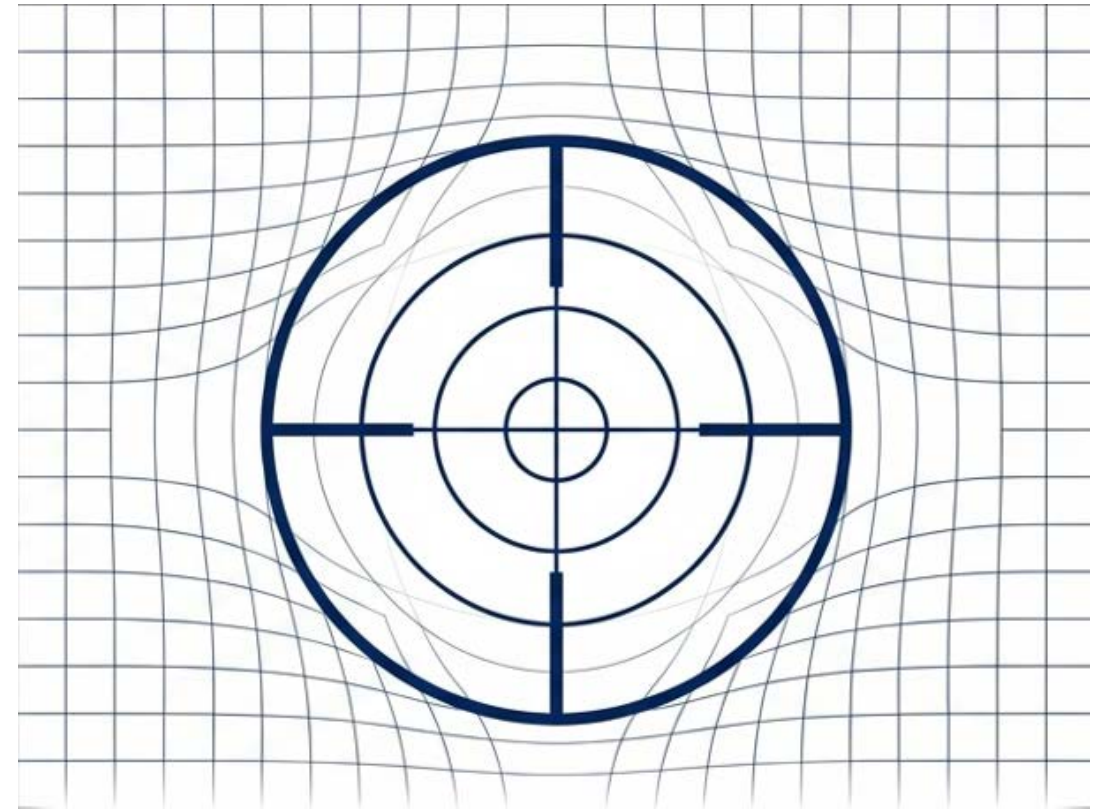
The 'Snapshot Fallacy' is obsolete. Patient-specific adaptation is required to safely manage daily anatomical shifts.

2 TECHNOLOGY HAS CLOSED THE GAP

Advances in AI, volumetric IGRT, and integrated platforms have moved ART from theoretical necessity to daily clinical reality.

3 THE FRONTIER IS BIOLOGICAL

The future of RT is moving beyond shape



The ultimate goal of Adaptive Radiotherapy:
Ensuring the dose delivered is exactly
the dose intended.